PSD-SQ: Point Set Decoding Based on Semantic Query for Object Detection in Remote Sensing Images

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***Abstract*— Object detection in remote sensing images (RSIs) remains a challenging task due to complex variations in object scale, dense arrangements, and arbitrary orientations. Compared to the widely used multistage and one-stage approaches, query-based methods that avoid postprocessing procedures and implement end-to-end inference, have recently attracted much attention. However, existing query-based methods still face two main challenges: 1) the feature sampling regions predicted by the query vectors often fail to be aligned with the foreground features, making it difficult to accurately classify and locate potential objects; and 2) the cascade decoders are crucial for optimizing the query vectors, resulting in a slower inference process. To address the above issues, we propose a novel object detection method named point set decoding based on semantic query (PSD-SQ), which mainly consists of two components: a semantic query generator (SQG) module and an oriented point set decoder (OPSD) module. The SQG module is proposed to generate semantic query vectors with rich object information based on the semantic correlations among feature vectors. The OPSD module includes two blocks: a point sampling with angle (PSA) block and a dynamic interactor (DI) block. The PSA block is constructed to refine the sampling locations with predicted angles, aligning the sampling locations and oriented object regions, and the DI block is designed to decode the sampled features with dynamic weights, making the decoding process more efficient. The proposed method is extensively evaluated on various object detection datasets of RSIs, and the experimental results consistently demonstrate that the proposed method achieves state-of-the-art (SOTA) performance in terms of both accuracy and inference speed. In addition, our code is available at https://github.com/I3ab/PSD_SQ.**

***Index Terms*— Object detection, point set representation, query-based model, remote sensing images (RSIs).**

I. Introduction

REMOTE sensing images (RSIs) are photographs of the Earth's surface captured by satellites or other aerial platforms, frequently employed for investigating land use, vegetation, and natural resources. Object detection in RSIs is still a challenging task that involves localizing and classifying objects in RSIs, and it has a wide range of applications including environmental monitoring, precision agriculture, and geographic information systems. Usually, objects in RSIs have the following characteristics.

1) Instances of the same class have arbitrary orientations, making it difficult to identify and classify them accurately.

2) Different instances exhibit significant scale variations, further complicating the task of precise recognition.

3) Most instances are densely arranged, as commonly observed in car parks and harbors, posing a challenge in distinguishing individual objects and their features. The characteristics mentioned above require the model to learn the exact locations of instance features and leverage multiple-scale feature information effectively to predict instance categories and locations.

With the rapid development of deep learning, object detection methods based on convolutional neural networks (CNNs) [1], [2], [3] and dense prediction have emerged and made significant progress in the field of object detection in RSIs. Several two-stage methods have been employed to perform object detection in RSIs with predefined anchors [4], [5], [6]. Following the common setting in two-stage methods, regions of interest (RoIs) are predicted from the horizontal anchors in the RoI-Transformer [4]. However, the horizontal anchors often misalign with the instance features, resulting in RoI features with excessive context information and making training more challenging. To mitigate this problem, RRPN [7] utilizes an oriented anchor setting to get a more accurate instance feature region. Unfortunately, the oriented anchor setting with angle parameters increases model complexity with tens of thousands of anchors and complex hyperparameters. Unlike relying on strong inductive biases for generating spatial priors, Oriented-RCNN [5] adopts predefined anchor points as the initial state of the bounding box regression and predicts class and oriented bounding box (OBB) offsets based on anchors' center features. Overall, the setting of oriented anchors in the two-stage methods results in complex hyperparameter tuning, and a large number of low-quality and redundant rotation proposals also limit the inference speed of the two-stage methods.

To avoid the complexity and hand-crafted anchors of two-stage methods, one-stage methods [2], [6], [8] are widely used. These methods rely on the refinement stage to reconstruct the feature map, allowing the feature map to contain

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relatively accurate semantic information. Based on the framework of RetinaNet [8], the feature maps are reconstructed by the oriented proposals' vertex and center point features in R3Det [9], [10]. Besides, S2ANet [11] designs a novel convolution operator to obtain refined features. Compared to the above refinement methods, the point set representation methods offer greater flexibility in capturing foreground features of objects [12]. Consequently, Oriented RepPoints [13] leverages point sets to represent instance features, achieving increased adaptability in the feature perception field. Since the objects in RSIs are oriented and their boundaries are irregularly distributed, the point set representation methods should be particularly beneficial for RSIs. In general, the local and axis-aligned receptive fields of ordinary CNN kernels may not match the foreground features of oriented objects, which makes it difficult for ordinary CNN features to accurately predict the position of sampling points.

Despite the promising performance of the methods mentioned above, the dense predictions and inaccurate feature sampling strategy hinder further improvements in model performance. Moreover, these methods often require a post-processing process, such as nonmaximum suppression (NMS) [14], [15] to filter prediction results.

Building on the success of the DETR [16], query-based methods [17], [18], [19] demonstrated state-of-the-art (SOTA) performance in generic object detection datasets. Usually, they represent potential instances as multiple learnable object queries. Attention layers for different query-based methods often have different structures and different strategies for initializing and updating object queries. Based on the framework of the two-stage Deformable-DETR [20], AO2-DETR [21] incorporates an oriented proposal generation network to predict oriented proposals, and the feature maps are refined by an adaptive oriented proposal refinement module. However, the query generation methods and feature sampling strategies employed in these query-based methods do not account for the unique characteristics of RSIs. Specifically, since the feature maps solely contain rudimentary semantic information about oriented objects, only a limited quantity of densely generated proposal boxes can depict the positions of the oriented objects. As a result, obtaining appropriate proposal boxes for query generation becomes a challenging task. Moreover, it is difficult to align oriented proposal boxes with the irregular object's edges, resulting in the RoI features incorporating excessive and irrelevant background information. These factors not only complicate the decoder structure, but also inhibit the performance of query-based methods.

To address the issues mentioned above, we propose a novel query-based object detection method named point set decoding based on the semantic query (PSD-SQ) in this article. Specifically, the proposed method mainly includes two components: a semantic query generator (SQG) module and an oriented point set decoder (OPSD) module. In the SQG module, the multihead self-attention (MHSA) [22] layer is employed to capture the semantic correlation among feature vectors. Since the foreground feature vector contains rich semantic information about objects and its correlation with other feature vectors is anticipated to be sparse, we propose the query score metric based on information entropy to measure the suitability of each feature vector for generating query vectors. The image features with higher query scores are retained for query generation. To decode query vectors and image features effectively and efficiently, we propose and design the OPSD module that consists of the PSA block and the DI block. The PSA block is leveraged to align the feature sampling positions with the foreground features of oriented objects, while the DI block is employed to decode the sampled feature. In the PSA block, the feature sampling positions are represented as sets of points. The initial coordinates and rotational angle of each point set are derived from the associated query vector. Subsequently, the final coordinates of a point set are obtained by applying the rotational angle to the initial coordinates. In the DI block, the dynamic decoding weights are generated from the query vectors. Then, these dynamic weights are used to decode the sampled features to obtain the object's category and position.

We conduct extensive experiments on various prevalent object detection datasets of RSIs. All experimental results consistently demonstrate the effectiveness of the proposed PSD-SQ in improving the accuracy of object detection in RSIs. Besides, the proposed method can also eliminate the necessity of adopting cascaded decoders, thus significantly improving the inference speed of query-based methods. Furthermore, we validate the effectiveness and generalization of each module in our method by conducting a series of ablation studies using different baseline models.

The main contributions of this article can be concisely summarized as follows.

1) To the best of our knowledge, the proposed PSD-SQ is the first query-based method that uses point set representation, unlike other query-based methods that use box regions for feature sampling. By employing point set representation, the query-based methods can more accurately sample instance features, achieving enhanced performance on the object detection task in RSIs.
2) We propose a query generator for query-based methods named SQG that can facilitate the prediction of accurate feature sampling locations, reducing the reliance of query-based models on multiple layers of decoders while ensuring optimal model performance.
3) In the proposed OPSD module, we construct the PSA block for point set refinement with a predicted angle and then design the DI block to decode the sampled feature dynamically, thereby achieving robust and effective feature decoding.

The remainder of this article is organized as follows: a comprehensive overview of the related works is given in Section II. In Section III, we present the proposed methods in detail. Section IV shows the experimental results and provides a clear and concise insightful analysis of the proposed method. Concluding remarks are presented in Section V.

II. Related Works

A. Object Detection in RSIs

Existing object detection methods used for RSIs are typically based on general object detection methods.

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To compensate for the domain differences between natural image pretraining models and RSIs, RSP [23] explores the impact of pretrained models on downstream tasks such as object detection and instance segmentation for RSIs.

In response to the arbitrary rotational angle of RSI instances, ReDet [24] explicitly represents the rotation-equivariance and rotation-invariance on feature maps, effectively mitigating the model's dependence on data augmentation. Since the arbitrary orientation of RSI instances also results in the exchange of edges (EOE) and periodicity of angles (POA) [25] problems in the OBB regression process, the discretized regression losses [26], [27], [28], [29] are adopted to mitigate the impact of EOE and POA problems. To avoid the occurrence of these two problems, GWD [30] and KLD [31] employ the parameters of a Gaussian distribution to express the regression box.

Therefore, OAN [32] improves this process by enabling detectors to focus on a smaller number of RSIs' patches, obtaining more efficient inference and more accurate results. Although OAN improves the model inference speed by reducing the number of inference samples, it has not addressed the issue of optimizing the model's computational complexity.

B. Query-Based Object Detection Methods

With the development of the transformer [22] and attention mechanism, query-based methods have achieved SOTA performance on multiple object detection datasets. Based on the vanilla transformer [22], DETR [16] uses a fixed set of small learning object queries to infer the relationship between the features of the objects and the global image context and directly outputs the final prediction set in parallel. Moreover, the attention weights in the transformer layer are considered to encompass substantial information regarding the semantics of objects. For example, TSP-FCOS and TSP-RCNN [33] show that the encoder's attention weights of specific points concentrate on the semantic regions of foreground instances, while the decoder's attention weights focus on the edges of these instances. Moreover, the attention weight implies the importance of each feature vector in PoWER-BERT [34], and only the important feature vectors in the shallow encoder are retained as inputs to the deep encoder.

To accelerate the convergence of query-based methods, Deformable-DETR [20] replaces self-attention with a linear attention mechanism using fully connected layers. Sparse RCNN [19] uses a dynamic head [35] to decode multiscale features, obtaining faster convergence and improved accuracy for object detection. Besides, the query tokens are critical in query-based methods. Anchor DETR [18] and DAB-DETR [36] reduce the training complexity of query-based methods by incorporating manually designed spatial priors to initialize query tokens. DN-DETR [37] inputs the ground-truth labels and boxes with added noise into the decoder and trains the model to reconstruct the original labels and boxes. DINO [38] improves the previous DETR-like models by using mixed query selection, contrastive denoising training, and a look-forward twice scheme. To better update queries, intermediate queries are collected cumulatively as the decoding stage is deepened and selectively forwarded to the downstream stage in SQR-DETR [39]. To achieve a balance between the detection performance and the number of transformer decoder layers, Efficient-DETR [40] takes the encoded features and utilizes RPN to generate proposals, which are then used as query token generation.

To overcome the difficulties arising from the arbitrary orientation and significant scale differences of instances in RSIs, AO2-DETR [21] utilizes Deformable-MHSA [20] to combine multiscale features in both encoding and decoding stages. Moreover, it aligns the features of oriented instances to facilitate object detection. Despite the remarkable achievements of query-based methods, their inference speeds are insufficient for offline deployment, especially for real-time analysis tasks in RSIs.

C. Feature Sampling in Oriented Object Detection

Due to the misalignment between the CNN features' axis-aligned receptive field and the oriented instance features, the image features need to be refined by feature sampling methods. According to the different instance feature sampling ways, existing oriented object detection methods can be mainly classified into three categories: RoI sampling, refinement process, and point sampling-based methods.

For the **RoI sampling-** based methods, the RoI-Transformer [4] predicts the rotated RoIs from horizontal RoIs. Subsequently, the detection head performs classification and regression on the features of rotated RoIs. Since horizontal RoIs usually contain too much background information, Oriented R-CNN [5] predicts the rotational offset of a horizontal anchor using the center-point feature of the anchor.

The **refinement process**-based methods are usually adopted in the one-stage methods to extract the features of oriented objects. The center point and vertex features of the predicted rotational box are reconstructed through bilinear interpolation in R3Det [9]. In S2ANet [11], the sampling range is obtained by predicting the rotational angle of the anchor, and the sampled features within the rotated anchors are then downsampled via parameterized convolutional layers.

For the **point sampling**-based methods, Oriented RepPoints [13] builds upon RepPoints [12] by rematching the feature points generated by DCN with instance labels and using the Jarvis March [41] algorithm to compute the convex hull of the point set. SASM [42] dynamically selects the positive sample range based on instance shape information and feature distribution. In CFC-Net [43], the point set obtained through prediction is utilized to construct a convex hull, and the features within the convex hull are used as the feature vectors for constructing candidate instances.

III. Proposed Method

Object detection of RSIs consists of two subtasks: object localization and object classification in RSIs. Specifically, given an image sample $x \in \mathbb{R}^{3\times H\times W}$, the backbone network F calculates its multilevel feature maps $f = F(x)$. Based on the feature maps $f \in \mathbb{R}^{D\times H^l\times W^l\times L}$, the classification and localization heads of the object detection model are used to

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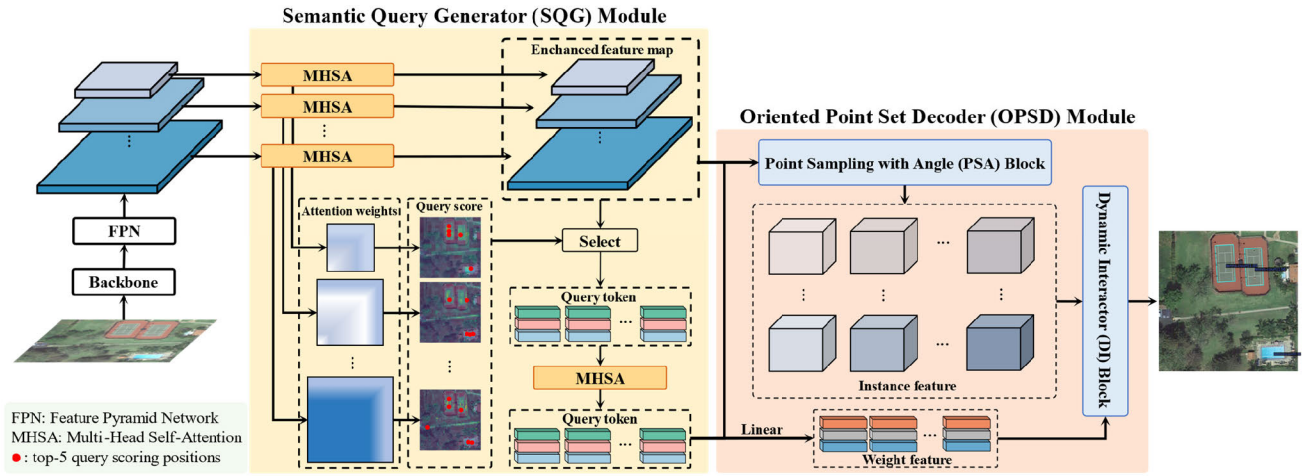


Fig. 1. Structure of PSD-SQ. The distribution of query scores is visualized on the image, where a brighter color indicates a higher score and a darker color corresponds to a lower score. Additionally, the top-5 scoring positions of each feature map are marked with red solid dots for better visibility.

calculate the oriented location $b = (x, y, w, h, \theta)$ and category c of each candidate object.

The vanilla MHSA is composed of three steps: linear transformation, similarity calculation, and weighted sum. In the linear transformation step, each feature vector on the feature map is mapped to the query Q, key K, and value V token spaces by different linear layers. The similarity between features, that is, attention weight, is obtained by the dot product of the query and key vectors. The value tokens V at different positions are weighted according to the attention weights and then aggregated by summing them up to obtain the output of MHSA. MHSA is defined as follows:

$$\text{MHSA}(Q,\ K,\ V) = \text{Concat}(h_1,\ \ldots,\ h_m,\ \ldots,\ h_M) \quad (1)$$

where m indexes the attention head. For each head h_m, the self-attention mechanism can be expressed as

$$h_m = \text{Attention}(Q_m, K_m, V_m) \quad (2)$$

$$Q_m = Q W_m^Q, \quad K_m = K W_m^K, \ V_m = V W_m^V \quad (3)$$

where $Q,\ K,\ V \in \mathbb{R}^D$, $W_m^{Q,\,\text{K,V}} \in \mathbb{R}^{D\times(D/M)}$, and the token dimension is set to be the same as the feature's dimension D as default. The self-attention mechanism for Q, K, and V tokens in each head is defined as follows:

$$\text{Attention}(Q_m, K_m, V_m) = \text{softmax}\left(\frac{Q_m K_m^\top}{\sqrt{\text{D}}}\right) V_m \quad (4)$$

where $K_m^\top$ denotes the transpose of K_m. The attention weight for each head is denoted as follows:

$$A_m = \text{softmax}\left(\frac{Q_m K_m^\top}{\sqrt{\text{D}}}\right). \quad (5)$$

Compared with randomly initialized or prior-based queries, the feature vectors enriched with comprehensive object semantic information enable more accurate querying of object locations. Indeed, the feature vectors enhanced by the MHSA layer encompass both the global receptive field and rich semantic information of objects. To better utilize this characteristic of MHSA, we should address two important issues: 1) how to determine which feature vectors contain rich instance information and 2) how to effectively decode the feature map by these feature vectors to obtain accurate regression and classification results.

To address both the above issues, we propose the PSD-SQ method and elaborate on the detailed structure of the proposed method in Fig. 1. The proposed PSD-SQ primarily comprises two modules: the SQG, which generates query vectors, and the OPSD, responsible for decoding. Within the SQG module, the image features are enhanced by MHSA. The query vectors are then obtained from the enhanced image features and the attention weights of the MHSA layers. Subsequently, the OPSD module employs the enhanced image features and the query vectors for decoding. Specifically, the object features are sampled from the enhanced image features in the PSA block. The sampled features are then dynamically decoded using the query vectors within the DI block.

Compared with other query-based methods, our PSD-SQ selects and maps the object query on the basis of the semantic relationship between feature vectors and refines the 2-D points' positions of feature sampling through the predicted rotation angle in terms of decoder structure. The detailed structure of each module is described below.

A. SQG Module

The SQG module generates query tokens for decoding based on the spatial and semantic relationships between different points on the feature map. It is mainly composed of three steps as follows.

1) Calculating the self-attention weights of the image features.

2) Computing the query score for each feature map based on attention weights.

3) Generating a specified number of query tokens based on the reconstructed feature map and the query score. The overall structure of the SQG module can be illustrated in Fig. 2.

To simplify the explanation, we select a one-layer feature map $f^l \in \mathbb{R}^{D\times H^l\times W^l}$ from the multilevel feature maps to illustrate the process of selecting feature vectors $f_{ij}^l \in \mathbb{R}^D$ as query tokens from the feature map. Then, we will explain

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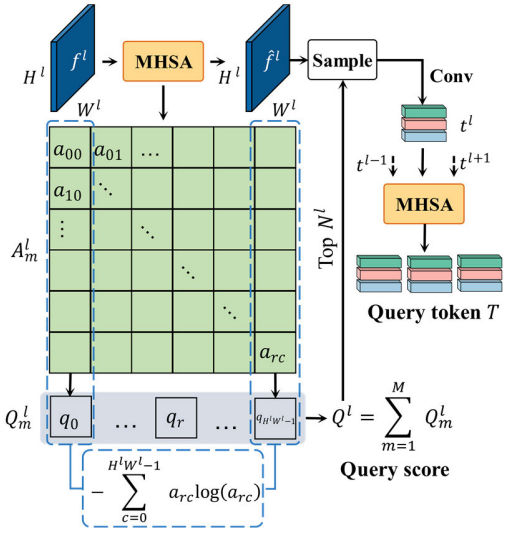


Fig. 2. Structure of the SQG module. The SQG module uses MHSA to capture the semantic correlation among feature points and calculates query scores to identify features suitable to generate the query vectors.

the process of fusing the query tokens generated from the multilevel feature maps. According to the definition of MHSA, the origin feature map f^l is transformed through MHSA to obtain the normalized attention weight $A^l \in \mathbb{R}^{M\times H^lW^l\times H^lW^l}$ and the reconstructed feature map $\hat{f}^l \in \mathbb{R}^{D\times H^l\times W^l}$:

$$A^l, \hat{f}^l = \mathrm{MHSA}\left(f^l\right). \tag{6}$$

For each head, the attention weight is represented as $A_m^l \in \mathbb{R}^{H^lW^l\times H^lW^l}$. It is worth noting that according to the calculation formula of MHSA, the SoftMax normalization of attention weight is performed along the last dimension of the matrix, which means that

$$\sum_{c=0}^{H^lW^l-1} A_m^l[r,c] = 1. \tag{7}$$

To calculate the query score for each feature point, we first calculate the negative information entropy for each column c of attention weights for each head $Q_m^l \in \mathbb{R}^{H^lW^l}$ as follows:

$$Q_m^l[r] = H\left(A_m^l[r,:]\right) \tag{8}$$

$$H\left(A_m^l[r,:]\right) = -\sum_{c=0}^{H^lW^l-1} a_{rc}\log(a_{rc}) \tag{9}$$

where $a_{rc} \geq 0$ and $\sum_{r=0}^{H''} a_{rc} = 1$. a_{rc} is the value at the rth row and cth column of the attention weight A_m^l. The query score $Q^l \in \mathbb{R}^{H'W'}$ of the feature map f^l is calculated as follows:

$$Q^l = \sum_{m=1}^{M} Q_m^l. \tag{10}$$

The top N^l feature vectors with the highest query scores of the feature map f^l is mapped to candidate query tokens $t^l \in \mathbb{R}^D$ by 1×1 Conv with the channel number of feature vectors, and the query token's origin position $o^l \in \mathbb{R}^2$ is normalized by the height and width of feature map f^l. We denote the set of query token candidates of multilevel feature maps as

$$T^l = \left\{t_0^l, t_1^l, \ldots, t_{N^l-1}^l\right\} \tag{11}$$

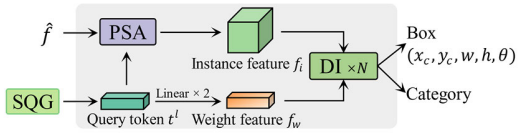


Fig. 3. Structure of the OPSD module. The OPSD module mainly consists of the PSA block and the DI block.

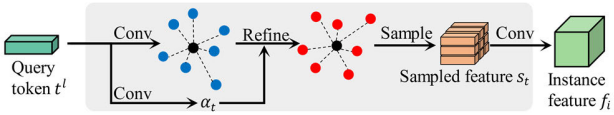


Fig. 4. Structure of the PSA block. PSA utilizes the rich instance information and maps the query tokens to rotational angles by means of a convolutional branch.

and its corresponding position as

$$O^l = \left\{o_0^l, o_1^l, \ldots, o_{N^l-1}^l\right\}. \tag{12}$$

All the query token candidates are preserved as the decoder's input, and the total number of query tokens is $N^l \times L$. Since some query tokens may originate from the same object instance, to reduce redundancy, we pass all the query tokens from each feature map through an MHSA layer to obtain the query tokens used for decoding.

B. OPSD Module

The OPSD module consists of the PSA block and the DI block. Initially, the query tokens from the SQG module are utilized to generate a point set for feature sampling in the PSA block. Subsequently, in the DI block, the sampled feature and the query token are mapped to the instance feature and weight feature, respectively, for dynamic decoding. The overall structure of the OPSD module can be shown in Fig. 3.

1) PSA Block: The feature map can be used to extract potential instance features through either point sampling or box sampling methods. We propose a new point sampling approach, namely PSA, which enables the decoder to accurately sample foreground features for oriented object detection tasks. The architecture of the PSA block can be detailed in Fig. 4.

For point sampling-based methods, the sampling positions for each instance are composed of a set of points. Unlike other point sampling methods, the PSA block utilizes predicted angles to perform the rotational transformations on the coordinates of points within a point set. A schematic of the different sampling methods can be shown in Fig. 5.

The PSA block takes query tokens T, their initial positions O, and enhanced multilevel feature maps $\hat{f}^l$ as input. It then produces sampled features, which are used to predict the positions of potential instances. Specifically, given N query tokens, each query token t is mapped to 2-D coordinate offsets relative to the original coordinate o_t through two linear layers. The original coordinate point o_t and the coordinate offsets are then combined to obtain the coordinates for feature sampling of the query token t

$$P_t = \{(x_1, y_1), \ldots, (x_h, y_h), \ldots, (x_H, y_H)\} \tag{13}$$

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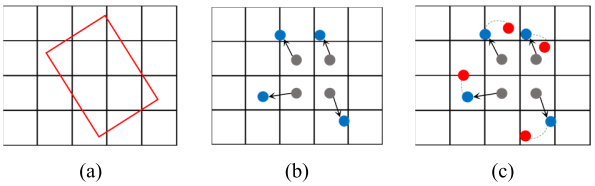


Fig. 5. Schematic of different sampling methods. (a) Oriented box sampling, which is typically adopted in Oriented RCNN [5], RoI-Transformer [4], and AO2-DETR [21]. (b) Point Sampling, which is typically used in Oriented RepPoints [13] and SASM [39]. (c) PSA (Ours), which utilizes a predicted angle to refine the sampling point set.

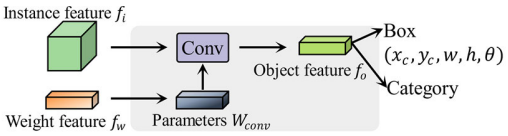


Fig. 6. Structure of the DI block. The DI block is a dynamic decoder for query-based methods.

where H is set to 9 as default in this article.

Previous sampling methods [12], [13], [42] usually rely solely on the flexibility of point sampling, thereby capturing foreground features by predicting the sampling positions based on the feature maps. However, since the receptive field of CNN features does not encompass angle information and instances in RSIs usually exhibit arbitrary rotational angles, these methods cannot attain precise foreground features.

Considering that the query tokens generated by the SQG module have a nonlocal receptive field, we propose to utilize the PSA block to map the query token to the rotational angles of sampling positions for oriented objects. Specifically, the query token t is mapped to a rotational angle α_t through two convolutional layers. Then, the points in P_t are rotated by α_t for feature sampling

$$\begin{cases} x_h^{'} = (x_h - x_c)\cos\alpha_t - (y_h - y_c)\sin\alpha_t + x_o \\ y_h^{'} = (x_h - x_c)\sin\alpha_t + (y_h - y_c)\cos\alpha_t + y_o. \end{cases} \tag{14}$$

The rotation center is defined as

$$x_c = \frac{\sum_{h=1}^{H} x_h}{H}, y_c = \frac{\sum_{h=1}^{H} y_h}{H}. \tag{15}$$

Thus, the rotated coordinates $P_t^{'}$ can be represented as

$$P_t^{'} = \left\{\left(x_1^{'}, y_1^{'}\right), \ldots, \left(x_h^{'}, y_h^{'}\right), \ldots, \left(x_H^{'}, y_H^{'}\right)\right\}. \tag{16}$$

Then, the sampled feature $s_t \in \mathbb{R}^{K\times D}$ is obtained by sampling from the enhanced feature map f'' using the refined coordinates of the point set. Since the coordinates $P_t^{'}$ are represented as floating-point values, the feature corresponding to each coordinate point is obtained using bilinear interpolation.

2) DI Block: Dynamic convolution is an efficient convolution operator that provides dedicated convolution weights for different samples, and it has been widely used in various downstream tasks such as object detection and segmentation. Considering the irregular edges and arbitrarily oriented instances in RSIs, we propose an object detection decoder named DI for query-based methods. The architecture of the DI block can be detailed in Fig. 6.

Specifically, we use the instance feature f_i obtained from the PSA block for both classification and regression purposes, and the feature utilized to generate dynamic convolution parameters is denoted as the weight feature f_w. The weight feature $f_w \in \mathbb{R}^{N\times K}$ is obtained by mapping the query token t through two linear layers. Subsequently, the resulting weight feature f_w is mapped into a standard 1×1 convolutional kernel's parameter, with the input and output channel numbers equal to the channel number of the instance feature f_i. After that, the instance feature f_i is convolved by the generated 1×1 convolutional kernel and activated by the ReLU function to obtain the object feature f_o. In the regression branch, the object feature f_o is first flattened, and then a three-layer linear network is employed to compute the final regression prediction. The three-layer linear network utilizes the ReLU activation function and has a hidden dimension D. The classification prediction is computed through one linear layer.

It is worth noting that, due to the high quality of the queries generated by the SQG module and their strong relevance with the foreground features in the input image, the DI block can achieve competitive performance without the need for a cascade structure. Additionally, having fewer decoder layers leads to a faster inference speed in query-based methods.

To achieve higher detection performance, we propose the DI block with a cascade structure. Within this structure, the object feature f_o is utilized as the query token t^l for the intermediate decoder in the subsequent stage. Instead of predicting bounding boxes for feature sampling, the intermediate decoder layers compute the set of sampling points in the PSA block. To calculate the regression loss of the intermediate DI block in the cascade structure, we define the oriented pseudo-boxes as the convex hull generated by the sampling point set of potential instances. Besides, the pseudo-boxes made up of points are not used for feature sampling, but solely for calculating the auxiliary regression loss of the intermediate layers. Furthermore, the auxiliary classification loss is also applied in the intermediate decoder layers.

C. Set Matching and Loss Function

Our model computes the set prediction loss based on a fixed number of predicted results, which allows us to update the model parameters effectively. The calculation of set prediction loss comprises two essential steps: set matching and loss calculation. First, the matching loss of each element in both the predicted and ground-truth sets is calculated based on their values, and subsequently, the Hungarian algorithm is used to find the matching with the minimum overall loss [16]. Then, based on the optimal matching result, the value of the current loss function for the model prediction is computed. The loss function is the same as the matching loss, but it solely calculates the loss value for the current pairing result. The loss function includes both classification loss $\mathcal{L}_{\text{cls}}$ and regression loss $\mathcal{L}_{\text{reg}}$

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{reg}}. \tag{17}$$

To ensure a fair comparison of the model performance with the existing methods, we choose Focal loss [8] as the

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classification loss $\mathcal{L}_{\text{cls}}$, and the localization loss $\mathcal{L}_{\text{reg}}$ is composed of the Smooth $L1$ loss and the Rotated intersection-over-union (IoU) loss

$$\mathcal{L}_{\text{cls}} = \lambda_{\text{Focal}}\mathcal{L}_{\text{Focal}} \tag{18}$$

$$\mathcal{L}_{\text{reg}} = \lambda_{\text{loc}}\mathcal{L}_{\text{loc}} + \lambda_{\text{RIoU}}\mathcal{L}_{\text{RIoU}}. \tag{19}$$

Let $y = \{y_i\}_{i=1}^{G}$ be the label of the sample image, where $y_i = (b_i, c_i)$ represents the bounding box b_i and its class label c_i. The predicted set of the model is denoted as $\hat{y} = \{\hat{y}_i\}_{i=1}^{N}$, where $\hat{y}_i = \left(\hat{b}_i, \hat{c}_i\right)$. The bounding box is represented using the OpenCV notation, and the rotational angle of the bounding box $\theta \in (0, \pi/2]$. For the smooth $L1$ loss, the calculation can be written as follows:

$$\mathcal{L}_{\text{loc}}\left(\hat{b}, b\right) = \sum\nolimits_{p\in\{x_c,y_c,w,h,\theta\}} \text{smooth}_{L_1}\left(\hat{b}_p - b_p\right). \tag{20}$$

Since an increase in IoU indicates a closer match between the predicted result and the label, the value of the Rotated IoU loss function is defined as $1 - \text{RIoU}$. The matching cost and loss function for cascade DI's intermediate layer is the same as the last layer.

IV. Experimental Results and Analyses

A. Datasets Description

Three public datasets are used to evaluate the performance of the proposed method, including DOTA-v1.0, HRSC2016, and UCAS-AOD, respectively.

The **DOTA** [44] datasets, which have been extensively utilized for evaluating performances of object detection methods in the context of RSIs, present a significant challenge due to their diverse range of object scales, orientations, and high instance density in certain images. The **DOTA-v1.0** dataset consists of 2806 large aerial images, with sizes ranging from approximately 800 × 800 to 4000 × 4000 pixels, and includes 188 282 instances across 15 common categories: plane (PL), baseball diamond (BD), bridge (BR), ground track field (GTF), small vehicle (SV), large vehicle (LV), ship (SH), tennis court (TC), basketball court (BC), storage tank (ST), soccer-ball field (SBF), roundabout (RA), harbor (HA), swimming pool (SP), and helicopter (HC).

The **HRSC2016** [45] dataset is a commonly used benchmark dataset for evaluating the performances of object detection methods in RSIs, specifically for detecting ships at the harbor. It comprises images of ships at the wharf, collected from six well-known harbors, and contains only one category, namely "ship." The image sizes range from 300 × 300 to 1500 × 900 pixels, and the dataset includes a total of 1061 images, with 436 for training, 181 for validation, and 444 for testing. During training, the training and validation sets are used, and random horizontal flipping is applied to augment the dataset.

UCAS-AOD [46] is an RSI object detection dataset that contains two types of targets, automobiles, and aircraft, as well as negative background samples. The dataset includes 1510 images, each with an approximate size of 659 × 1280 pixels, and a total of 14 596 instances. For the experiments conducted in this study, the dataset was split into 1057 randomly selected images for training and 302 images for testing.

B. Experimental Setup and Evaluation Metrics

1) Experimental Setup: We implemented the proposed PSD-SQ based on the MMRotate [47] framework. All evaluation metrics are aligned with the MMRotate benchmark. Without explicitly specifying, all experiments on the DOTA dataset were conducted for the 3× scheduler. For the HRSC2016 and UCAS-AOD datasets, the training scheduler is set to 6×. The model training was performed using 4 NVIDIA RTX A6000 GPU devices, each with 48G memory. The total batch size used for training was set to 4. The model inference is completed on a single NVIDIA RTX A6000 GPU, and the batch size is set to 1. We trained the model with an initial learning rate of 1e-4. The learning rate is reduced by a factor of 10 at the 24th and 34th epochs. We use linear warmup for 500 iterations and the warmup ratio is set to 1/3. The weights of the loss function consist of three parts, and unless otherwise stated, these weight values are set to $\lambda_{\text{Focal}} = 5, \lambda_{\text{loc}} = 2$, and $\lambda_{\text{RIoU}} = 3$, respectively. We use the multilevel features $\{P3, P4, P5\}$ output by the standard FPN [48] as input for the SQG module. The number of query tokens generated on each feature map is set to 50. Therefore, the total number of query tokens is 150. In the case of a multilayer DI decoder, the auxiliary loss settings for intermediate decoder layers are the same as the last DI decoder. In addition to the commonly used ResNet18, ResNet50, and ResNet101 backbones, we also train the model using the transformer-based [23] pretrained on MillionAID [49] for better performance.

2) Evaluation Metrics: For quantitative evaluation, the mean average precision (mAP) provides a comprehensive assessment of the object detection model's accuracy by considering precision and recall at different confidence thresholds. To compute mAP, the model generates bounding boxes with confidence scores for detected objects, and the precision–recall curve is calculated for each object class based on ground-truth annotations. The AP, representing the area under the precision–recall curve, is then averaged across all classes to derive mAP. For the DOTAv1.0 dataset, the PASCAL VOC07 metric [50] is adopted. This metric focuses on evaluating the model's performance at a specific IoU threshold of 50%, which measures the overlap between predicted and ground-truth bounding boxes. For HRSC2016 and UCAS-AOD datasets, we use the PASCAL VOC 07 metric (11-point interpolated average precision) and the PASCAL VOC 12 metric (AUC method by integrating the precision–recall curve) [51].

Frames per second (FPS) is a vital metric for assessing real-time performance. High FPS values are crucial for prompt and efficient object identification in dynamic environments. The FPS metric is evaluated on the DOTAv1.0 dataset.

C. Experimental Results

We conducted a comparative evaluation of the proposed PSD-SQ with several SOTA models on multiple object detection datasets of RSIs. The experimental results are shown in Tables I–III. It is worth noting that due to the small size and small number of categories of the HRSC and UCAS-AOD datasets, not all methods have been experimented on these two datasets, making it difficult to obtain their

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TABLE I

COMPARISON OF PSD-SQ WITH SOTA METHODS ON THE DOTA-V1.0 OBB TASK. THE † SYMBOL INDICATES MULTISCALE TRAINING AND TESTING. THE ∗ SYMBOL INDICATES THAT THE FPS IS EVALUATED USING A SINGLE NVIDIA RTX A6000 GPU. THE RESULTS IN BOLD DENOTE THE BEST RESULTS IN EACH COLUMN, AND THE ITALIC TEXT INDICATES THE SECOND-BEST RESULTS

	Method	Backbone	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP	FPS*
Two-stage	Faster R-CNN-O [1] (2015)	R-50-FPN	88.44	73.06	44.86	59.09	73.25	71.49	77.11	90.84	78.94	83.90	48.59	62.95	62.18	64.91	56.18	69.05	16.8
	RoI Transformer [4] (2019)	R-101-FPN	88.65	82.60	52.53	70.87	77.93	76.67	86.87	90.71	83.83	82.51	53.95	67.61	74.67	68.75	61.03	74.61	15.7
	SCRDet [10] (2019)	R-101-FPN	**89.98**	80.65	52.09	68.36	68.36	60.32	72.41	90.85	87.94	**86.86**	65.02	66.68	66.25	68.24	65.21	72.61	/
	Gliding Vertex [26] (2021)	R-101-FPN	89.64	85.00	52.26	77.34	73.01	73.14	86.82	90.74	79.02	*86.81*	59.55	**70.91**	72.94	70.86	57.32	73.23	16.4
	CenterMap [29] (2021)	R-50-FPN	88.88	81.24	53.15	60.65	78.62	66.55	78.10	88.83	77.80	83.61	49.36	66.19	72.10	72.36	58.70	71.74	/
	ReDet [24] (2021)	Re50-ReFPN	88.79	82.64	53.97	74.00	78.13	**84.06**	88.04	90.89	87.78	85.75	61.76	60.39	75.96	68.07	63.59	76.25	11.2
	Oriented-RCNN [5] (2021)	R-50-FPN	88.86	83.48	55.27	76.92	74.27	82.10	87.52	*90.90*	85.56	85.33	65.51	66.82	74.36	70.15	57.28	76.28	16.7
One-stage	RetinaNet-O [8] (2017)	R-50-FPN	88.67	77.62	41.81	58.17	74.58	71.64	79.11	90.29	82.18	74.32	54.75	60.60	62.57	69.67	60.64	68.43	17.2
	S2ANet [11] (2022)	R-50-FPN	89.11	82.84	48.37	71.11	78.11	78.39	87.25	90.83	84.90	85.64	60.36	62.60	65.26	69.13	57.94	74.12	15.8
	R3Det [9] (2021)	R-101-FPN	88.76	83.09	50.91	67.27	76.23	80.39	86.72	90.78	84.68	83.24	61.98	61.35	66.91	70.63	53.94	73.79	12.8
	RSDet [27] (2021)	R-101-FPN	89.80	82.90	48.60	65.20	69.50	70.10	70.20	90.50	85.60	83.40	62.50	63.90	65.60	67.20	68.00	72.20	/
	KLD [30] (2021)	R-50-FPN	88.91	85.23	53.64	**81.23**	78.20	76.99	84.58	89.50	86.84	86.38	*71.69*	68.06	75.95	72.23	*75.42*	78.32	16.1
	SASM [42] (2022)	R-50-FPN	86.42	78.97	52.47	69.84	77.30	75.99	86.72	90.89	82.63	85.66	60.13	68.25	73.98	72.22	62.37	74.92	15.8
	Oriented-RepPoints [13] (2022)	R-50-FPN	87.02	83.17	54.13	71.16	80.18	78.40	87.28	*90.90*	85.97	86.25	59.90	*70.49*	73.53	72.27	58.97	75.79	16.7
Query-based	Rotated D-DETR [20] (2021)	R-50-FPN	75.87	56.79	17.38	39.14	36.31	13.79	19.39	87.65	66.54	48.81	18.99	47.59	29.30	52.73	17.58	41.86	/
	AO2-DETR [21] (2023)	R-50-FPN	89.27	84.97	56.67	74.89	78.87	82.73	87.35	90.50	84.68	85.41	61.97	69.96	74.68	*72.39*	71.62	77.73	14.2
	PSD-SQ (ours)	R18-FPN	87.04	81.32	53.79	69.32	75.38	77.21	87.98	88.96	75.17	84.26	68.97	62.43	68.61	68.55	62.74	74.12	**31.3**
		R50-FPN	89.67	85.43	57.28	75.24	79.98	81.21	88.30	89.88	*88.05*	86.35	69.63	68.45	75.32	69.46	72.65	78.46	*24.2*
		ViTAE(RSP)-FPN	*89.84*	*86.32*	**57.83**	75.64	*80.86*	81.87	*88.91*	90.21	**88.92**	86.80	70.84	69.55	*76.89*	70.79	72.86	*79.21*	7.6
	PSD-SQ† (ours)	R50-FPN	89.79	**86.87**	*57.78*	*78.97*	**80.95**	*82.98*	**89.94**	**91.67**	87.69	85.87	**72.93**	69.76	**80.97**	**72.89**	**73.94**	**80.20**	/

TABLE II

RESULTS ON HRSC2016. TEST SETS MAP(07) AND MAP(12) REPRESENT THE RESULTS UNDER VOC2007 AND VOC2012 MAP METRICS, RESPECTIVELY

Method	Backbone	mAP(07)	mAP(12)
RoI-Transformer [4]	R-101-FPN	86.20	/
S2ANet [11]	R-101-FPN	90.17	95.01
Oriented-RCNN [5]	R-50-FPN	**90.40**	96.50
AO2-DETR [21]	R-50-FPN	88.12	97.47
PSD-SQ (Ours)	R-50-FPN	90.23	**98.12**

TABLE III

RESULTS ON UCAS-AOD

Method	Backbone	mAP(07)
RetinaNet-O [8]	R-101-FPN	87.57
RoI-Transformer [4]	R-101-FPN	88.95
DAL [6]	R-50-FPN	89.87
Oriented RepPoints [13]	R-50-FPN	90.11
PSD-SQ (Ours)	R-50-FPN	**91.13**

corresponding experimental results or training details. As a result, the performance of only a few methods is listed in Tables II and III.

Results on DOTA-v1.0: We present the complete experimental results using both single-scale and multiscale samples on the DOTA dataset for fair comparison with previous methods. Using R-50-FPN as backbone networks, the proposed PSD-SQ achieves mAP scores of 78.46%. It obtains higher mAP scores using an RSI pretrained ViTAE backbone and also achieves higher FPS using a lightweight R-18-FPN backbone. Table I displays the test results of the model on DOTA-v1.0. Besides, several samples are chosen from the DOTA dataset to visualize the prediction results. The visualization results are displayed in Fig. 7.

It should be noted that FPS results are mainly affected by model structure, image size, and code efficiency. For PSD-SQ†(ours), †means that the size of the training/test images is variable, so the FPS results for PSD-SQ†(ours) are difficult to fairly compare with the other methods in Table I. As for SCRDet [10], Rotated D-DETR [20], ReDet [24], RSDet [27], and CenterMap [29], these methods are all implemented based on different deep-learning frameworks, and their FPS evaluation codes are either unavailable or difficult to unify, and the efficiency of CUDA operators cannot be consistent with that of MMRotate framework, making it difficult to make fair comparisons.

Moreover, the speed difference between the one-stage and two-stage methods used for object detection in RSIs is not significant because a large number of oriented anchors need to be set up in the anchor-based one-stage methods [4], [5], and the anchor-free one-stage methods need to include the refinement stages [9], [11] to reconstruct the feature map.

Results on HRSC2016: The HRSC2016 dataset is a ship detection dataset, with some ship instances having large aspect ratios and various orientations. We report the results using both VOC2007 and VOC2012 metrics. As shown in Table II, the proposed PSD-SQ achieves the best performance under both VOC2012 and VOC2007 metrics using the ResNet-50-FPN backbone without the need for a more complicated architecture.

Results on UCAS-AOD: UCAS-AOD datasets contain a large number of small objects with complex surrounding scenes. Table III shows the evaluation results with the recent methods on the UCAS-AOD dataset. Our proposed method achieves the best performance of 91.13% mAP.

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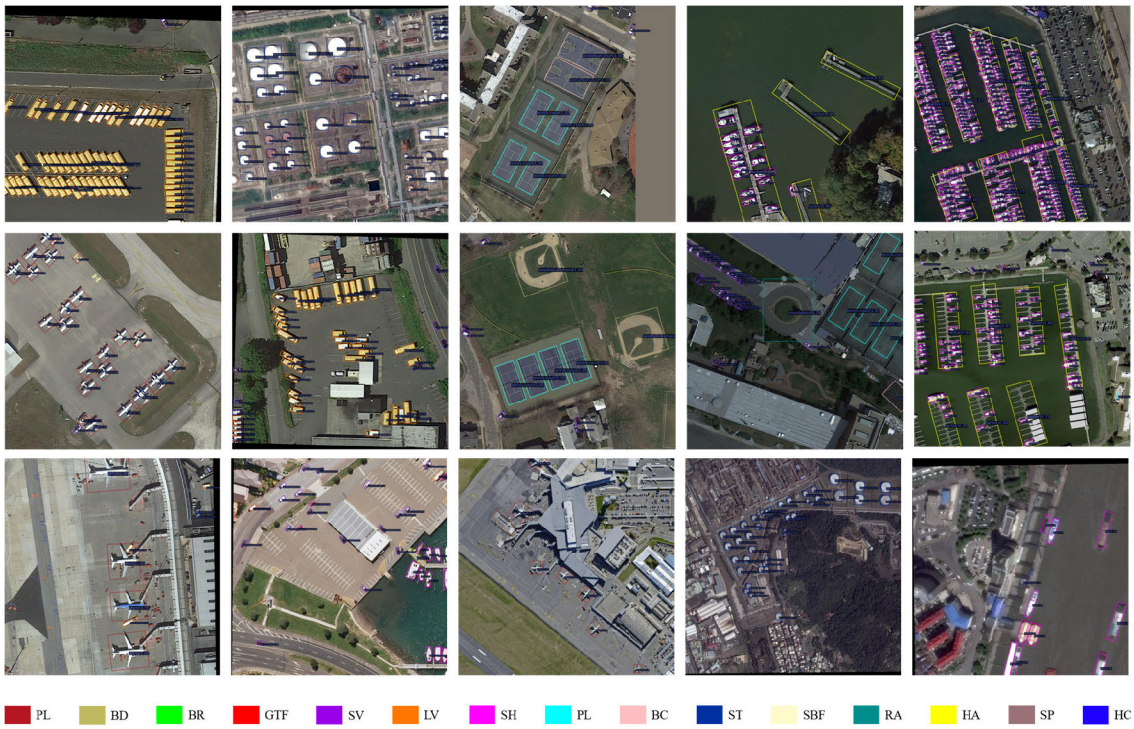


Fig. 7. Examples of detection results on the DOTA dataset using PSD-SQ with one-stage decoder and R-50-FPN backbone. The confidence threshold is set to 0.3 when visualizing these results. One color stands for one object class.

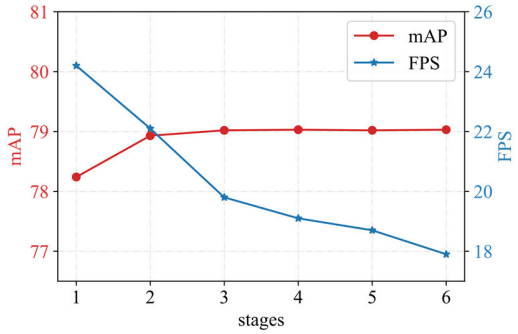


Fig. 8. mAP and FPS metrics vary with the number of layers in the decoder.

TABLE IV

EFFECT OF THE SQG MODULE AND THE PSA BLOCK. THE MAP IS REPORTED ON DOTA-V1.0. ∗ INDICATES THAT THE FEATURE MAPS USED FOR DECODING ARE NOT ENHANCED BY MHSA LAYERS, WHICH MEANS THAT THE SQG MODULE ARE ONLY USED FOR QUERY GENERATION

Method	SQG	PSA	mAP
PSD-SQ	-	-	72.65
	-	√	73.13
	√	-	77.92
	√	√	78.24
PSD-SQ*	√	√	76.18

D. Ablation Study

In this section, we conducted a series of ablation experiments on the DOTA-v1.0 test set to validate the effectiveness of our method. In all experiments, we use R-50-FPN as the backbone network.

Fig. 8 shows the impact of the number of cascade stages on the model's performance. It can be seen that compared to other query-based methods [16], [17], [19], [20], [21], the detection accuracy (mAP) of the proposed PSD-SQ remains basically unchanged, while the inference speed (FPS) decreases with the increase of the number of cascade stages.

Table IV shows the performance changes as we gradually added the SQG module and the PSA block to the model. The experimental results in Table IV also highlight the performance enhancement achieved by the high-quality query generated by SQG and PSA in OPSD. Besides, the MHSA layers in PSD-SQ∗ do not enhance the feature maps used for decoding, which means that the SQG module is only used for query generation. The performance gap indicates that the enhancements are better suited to the decoding process.

Table V shows the differences between PSD-SQ and other widely used query-based methods in terms of query generation, and its experimental results verify the effectiveness of the SQG module. Since the decoder design of different methods usually aims at different downstream tasks, such as RSIs or natural images, and contains many details and tricks, the OPSD module may not be directly applied to other methods.

E. Effectiveness and Generalization of the PSA Block

To further analyze the role of the PSA block, we visualize the output of the PSA block in our PSD-SQ method. In Fig. 9, the initial positions of sampling point sets are colored in blue. Besides, the red points are used to represent the positions of the sample points after the rotational transformation. As can

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Fig. 9. Visualization of the PSA block's prediction. Blue points are the initial sampling points, and the red points are the refined sampling points.

TABLE V
MAIN DIFFERENCE BETWEEN PSD-SQ AND OTHER WIDELY USED QUERY-BASED METHODS IN TERMS OF QUERY GENERATION. THE MAP IS REPORTED ON DOTA-V1.0

Baseline	Query Generator	mAP
Rotated Deformable-DETR [20]	Randomly Initialized	41.86
	SQG	**43.87**
Rotated SparseRCNN [19]	Randomly Initialized	76.54
	SQG	**77.23**

TABLE VI
GENERALIZATION OF THE PSA BLOCK. THE MAP IS REPORTED ON THE DOTA-V1.0 TEST SET

Method	PSA	Angle	Batch Size	mAP
Oriented RepPoints [13]	-	le135	2	71.94
	√			**72.13**
SASM [42]	-	le135	2	66.45
	√			**67.72**

be observed from the figure, for various object types, there is a higher consistency between the positions of the sampled points after rotation correction and the foreground ranges of the objects.

To validate the generalization of the PSA module, we used Oriented RepPoints [13] and SASM [42] as baseline models and tested them on the DOTA-v1.0 test set. The test results are shown in Table VI.

F. Advantages of Point Set Representation-Based Methods

To further illustrate the advantage of the point set representation-based methods, we conducted a series of ablation experiments, the results of which are shown in Table VII. For the oriented-anchor-based methods, we select Faster-RCNN-O [1] as the baseline. With the increase in the number of oriented anchors, the model's mAP gradually increases, but the inference speed decreases. For the point set representation-based methods, the experimental results of Oriented-RepPoints [13] and our PSD-SQ are provided. With the change in the number of sampling points, the inference speed of the model remains basically unchanged. When the number of sampling points increases, the AP increases slightly. Overall, the oriented-anchor-based methods are more sensitive to hyperparameters and have slower inference speeds, while the point set representation-based methods have simpler hyperparameter tuning and faster inference speeds.

TABLE VII
PERFORMANCE COMPARISON OF THE ORIENTED-ANCHOR-BASED METHOD AND THE POINT SET REPRESENTATION-BASED METHOD. "1 × 4 × /" INDICATES THAT THE ANCHOR IS INITIALIZED WITH THE HORIZONTAL ANCHOR. "H = 5/9/13" INDICATES THAT THE NUMBER OF SAMPLING POINTS IS SET TO 9

Method	Number of Anchor (scale × ratio × angle)	mAP	FPS
Faster-RCNN-O [1]	1 × 4 × /	69.05	16.8
	1 × 4 × 4	72.32	15.1
	1 × 4 × 6	73.14	14.3
	3 × 4 × 6	73.53	12.4
Oriented RepPoints [13]	H = 5	76.02	16.8
	H = 9	76.28	16.7
	H = 13	76.25	16.7
PSD-SQ (Ours)	H=5	78.42	**24.4**
	H = 9	**78.46**	24.2
	H = 13	78.51	24.2

G. Insightful Analyses

By optimizing query generation by means of the SQG module and improving the decoding process by using the OPSD module, the proposed PSD-SQ has achieved significant improvements in both accuracy and inference speed for object detection tasks in RSIs. Based on the above experimental results, we provide the following insightful analyses.

1) Since the self-attention mechanism in MHSA can capture the relationship between feature vectors, the semantic query vectors generated by the SQG module eliminate the need for cascade decoders used to obtain high-quality query vectors from randomly initialized query vectors. Moreover, these semantic query vectors not only simplify the decoder structure and accelerate the inference speed, but also provide sufficient support for the sampling process of point set features.
2) In the OPSD module, the PSA block effectively utilizes the semantic information in query vectors to refine the sampling point set, achieving better alignment between the sampling position and the foreground features of the oriented object. Moreover, as a plug-and-play block, PSA can significantly improve the process of feature sampling for oriented instances in the point set representation methods.
3) The DI block uses dynamic weights to decode the point set representation corresponding to each query, which improves the generalization and robustness of the decoder and ensures detection accuracy. To further improve the detection accuracy, we also propose the

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OPSD module with cascade DI block, however, the cascading of DI blocks makes the decoder more complex.

4) Since PSD-SQ is an end-to-end query-based model, both the training and inference processes are concise and efficient. The model utilizes commonly used loss functions, making hyperparameter tuning a straightforward task.

Since the number of objects in RSIs is dynamic and unpredictable, using a fixed number of query vectors may not be sufficient to detect all instances, especially when there are too many potential objects. Conversely, when there are only a few potential objects, the proposed method may still perform unnecessary decoding operations. Therefore, our future research will focus on developing a dynamic method for adjusting the number of query vectors to address this challenge effectively.

V. Conclusion

In this article, we have proposed PSD-SQ, a novel method for object detection in RSIs. PSD-SQ is a simple, end-to-end, and query-based framework that overcomes the limitations of previous query-based methods, that is, the reliance on cascade decoders and the inability to align sample positions with oriented objects. The proposed query generator, SQG, enables the generation of query vectors that accurately match the foreground features of objects, eliminating the need for cascade decoders and simplifying the entire process while enhancing efficiency. The OPSD module utilizes the PSA block to refine the sampling point set, aligning it better with foreground features. Moreover, the PSA block demonstrates flexibility, showing its potential to improve the performance of various point set representation methods. Besides, the DI block uses dynamic weights to decode the point set representation, improving the generalization and robustness of the proposed method.

Since the variability in the number of objects poses a challenge when using a fixed number of query vectors, our future work may focus on how to further optimize the object detection process in RSIs, such as developing a method to dynamically adjust the number of query vectors, which may reduce the redundancy of inference results and further improve inference efficiency.

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